Calibration of Probabilities

Assignment 2 — Experimental Project

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Statistical Methods for Machine Learning
Università degli Studi di Milano — A.Y. 2025/26

August 25, 2026

Code and results: `https://github.com/Behradsadeghi/ml-calibration-project`

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1 Introduction

Supervised learning is usually presented through risk minimisation: a model is trained to minimise a surrogate loss, and is then judged by classification accuracy. In many applications this is not enough. A medical triage system that outputs 0.8 should be right about 80% of the time it says 0.8; otherwise a downstream decision rule that thresholds on cost cannot be trusted. A model with this property is called *calibrated*, and calibration is a genuinely different property from accuracy: because Platt scaling and isotonic regression are monotonic transformations of the score, they leave the ranking (and therefore AUC) untouched while changing the probabilities substantially.

This project reproduces, on a small scale, the study of Niculescu-Mizil and Caruana [1]. Two off-the-shelf classifiers (Logistic Regression and Random Forest) are trained on two real-world datasets; their raw probability estimates are then post-processed with two calibration methods — Platt scaling and isotonic regression — both implemented from scratch. Results are evaluated with accuracy, log-loss and Brier score, and visualised with reliability diagrams.

The main contributions of this report, beyond the required experiments, are: (i) a multi-seed robustness check showing that several conclusions drawn from a single split do not survive repetition; (ii) a signed statistic that identifies the *direction* of miscalibration, not only its magnitude; and (iii) an honest account of a case where the calibration methods did not help, and of why that is the expected outcome for these two model families.

2 Datasets and why they were chosen

The assignment requires two real-world classification datasets and a comparison across them. I chose two that differ in the two properties that most directly drive calibration behaviour: how separable the classes are, and how noisy the labels are.

Breast Cancer is an "easy", well-separated problem, so I expected both models to start out close to calibrated: it serves as a low-miscalibration reference point. Diabetes is noisier and less separable, so I expected more miscalibration and more room for the calibration methods to help.

	Breast Cancer Wisconsin	**Pima Indians Diabetes**
Source	`sklearn.datasets`	OpenML `data_id=37` (UCI mirror)
Samples / features	569 / 30	768 / 8
Positive rate	$\approx 37\%$ (malignant)	$\approx 35\%$ (tested positive)
Separability	high ($\approx 97\%$ test accuracy)	moderate ($\approx 76\%$ test accuracy)

Table 1: The two datasets. Both are binary and small-to-medium scale, as required.

As Section 7 shows, that second expectation was only partly borne out — which is itself an informative result.

The Diabetes data uses 0 as a placeholder for missing measurements in five clinical features (glucose, blood pressure, skin thickness, insulin, BMI), which is not physiologically valid. These zeros are treated as missing and median-imputed using statistics computed on the training split only.

3 Methodology

3.1 Splitting protocol

Each dataset is split into three disjoint, stratified parts: 60% training, 20% calibration, 20% test. Stratification preserves the positive rate across all three parts, which matters for the imbalanced Diabetes data.

The separate calibration split is essential rather than cosmetic. Platt scaling and isotonic regression are themselves fitted to data. Fitting them on the training split would correct the base model's *training-set* overconfidence rather than its generalisation behaviour; fitting them on the test split would leak test labels into the model and inflate every reported number. Only a third, held-out split gives an honest estimate. The same reasoning applies to feature standardisation and to median imputation, both of which are fitted on the training split alone.

3.2 Base models

Logistic Regression and Random Forest (300 trees) are used off-the-shelf from scikit-learn, as the assignment permits. Their hyperparameters are not the object of study: they are used purely as sources of imperfectly calibrated probabilities. Both calibrators are fitted on top of each model's raw `predict_proba` output rather than a decision function, so that the two methods are applied identically to both models even though a Random Forest has no natural margin. I tested the alternative (applying Platt scaling to the log-odds of the probability) and it was clearly worse on Breast Cancer (mean log-loss 0.108 vs. 0.086 for Logistic Regression) and marginally better on Diabetes, so the simpler choice was kept.

3.3 Platt scaling (from scratch)

Platt scaling fits a two-parameter sigmoid on top of the raw score f:

$$P(y = 1 \mid f) = \frac{1}{1 + \exp(Af + B)}. \quad (1)$$

The parameters (A, B) are chosen by maximum likelihood, which is convex, and are found with a hand-written Newton's method (closed-form inverse of the 2×2 Hessian). Only two parameters are fitted, which makes the method extremely data-efficient but restricts it to monotonic, sigmoid-shaped corrections.

Crucially, the regression targets are not the raw labels $\{0,1\}$ but the Bayesian-smoothed values

$$t_i = \frac{N_+ + 1}{N_+ + 2} \text{ if } y_i = 1, \qquad t_i = \frac{1}{N_- + 2} \text{ if } y_i = 0, \tag{2}$$

where N_+ and N_- count positives and negatives in the calibration set. Without this smoothing, perfectly separable calibration data would drive $|A| \to \infty$ and collapse the sigmoid into a step function. With it, the fitted probability can never reach exactly 0 or 1 — a property that turns out to matter a great deal in Section 4.

3.4 Isotonic regression (from scratch)

Isotonic regression solves

$$\min_{g \text{ non-decreasing}} \sum_i (y_i - g(f_i))^2 \tag{3}$$

using the Pool-Adjacent-Violators Algorithm (PAVA), implemented as a single stack-based pass in $O(n)$ amortised time: points are sorted by score, each starts as its own block, and adjacent blocks are merged whenever the left block's mean exceeds the right block's, re-checking backwards after each merge.

Being non-parametric, isotonic regression can represent *any* monotonic correction, including shapes a sigmoid cannot express — but it has far more effective degrees of freedom and therefore needs more calibration data.

For prediction I interpolate linearly between the block means rather than returning a hard step function. This is a deliberate smoothing choice and I state it explicitly: the exactness guarantee of PAVA applies to the fitted blocks, not to what interpolation returns, so predictions at training points are slightly worse in residual sum of squares than the exact step solution. Scikit-learn interpolates between block *boundaries* instead and does reproduce the exact values. Both are monotonic; the difference is small.

3.5 Metrics and visualisation

Accuracy, log-loss, Brier score, reliability-diagram binning and Expected Calibration Error (ECE) are all implemented from scratch with numpy. Log-loss clips probabilities to $[\varepsilon, 1-\varepsilon]$ with $\varepsilon = 10^{-15}$, since an unclipped confidently-wrong prediction would make it infinite.

The three required metrics measure different things, which is why they can disagree. Accuracy sees only the hard decision at threshold 0.5 and is blind to confidence. Log-loss and Brier score are both proper scoring rules, but log-loss is unbounded near $p \in \{0,1\}$ while Brier score is bounded in $[0,1]$ and penalises extreme errors only quadratically. ECE, reported as a bonus, is the sample-weighted average vertical gap between the reliability curve and the diagonal.

Reliability diagrams use ten equal-width bins and are paired with a histogram of predicted probabilities underneath, following [1]: a curve computed from a nearly empty bin is noisy, and the histogram shows how much of the curve to trust. The binning implementation was cross-checked against `sklearn.calibration.calibration_curve` and agrees to $\approx 10^{-16}$.

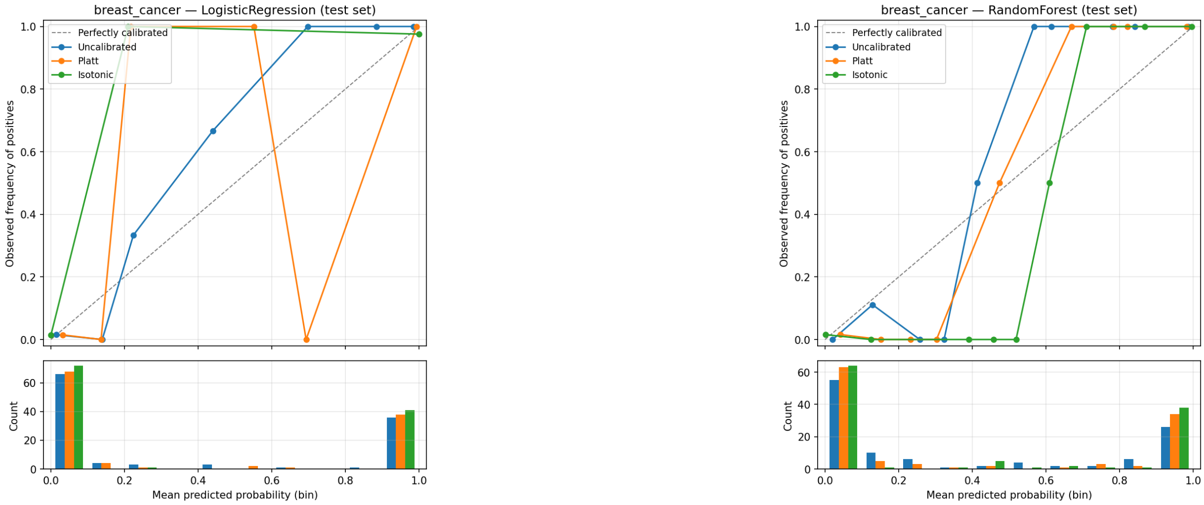
4 Results

4.1 A degenerate case for isotonic regression

The most striking number in Table 2 is the log-loss of 0.6196 for isotonic regression on Breast Cancer with Logistic Regression — seven times worse than the uncalibrated baseline — while Brier score ($0.0239 \to 0.0230$) and accuracy ($0.965 \to 0.974$) barely move, and in fact slightly improve.

Dataset	Model / method	Accuracy	Log-loss	Brier	ECE
Breast Cancer	LR — uncalibrated	0.9649	0.0861	0.0239	0.0220
	LR — Platt	0.9737	0.0885	0.0226	0.0381
	LR — isotonic	0.9737	0.6196	0.0230	0.0245
	RF — uncalibrated	0.9825	0.1177	0.0296	0.0672
	RF — Platt	0.9825	**0.0947**	**0.0193**	0.0468
	RF — isotonic	0.9474	0.3640	0.0293	0.0455
Diabetes	LR — uncalibrated	0.7403	0.5179	0.1743	0.0797
	LR — Platt	0.7468	0.5241	0.1750	0.0528
	LR — isotonic	0.6753	1.5688	0.1896	0.1272
	RF — uncalibrated	0.7208	0.5156	0.1751	0.0645
	RF — Platt	0.7208	0.5227	0.1761	0.0654
	RF — isotonic	0.7338	0.5193	0.1760	0.0403

Table 2: Test-set metrics for the single reference split (`SEED = 42`). Lower is better for log-loss, Brier and ECE.



(a) Breast Cancer, LR

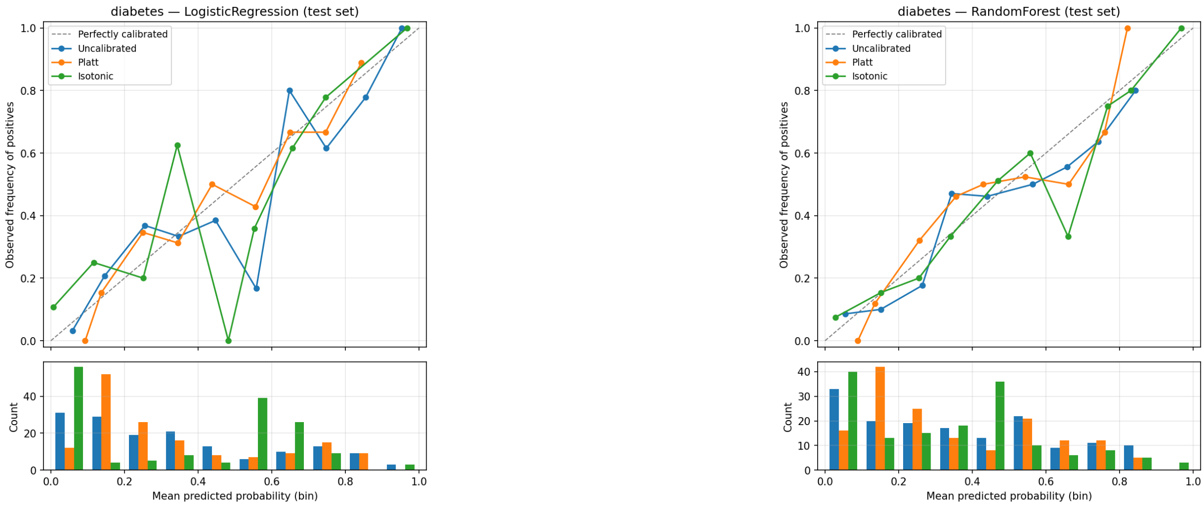
(b) Breast Cancer, RF

Figure 1: Reliability diagrams before and after correction, with the histogram of predicted probabilities below each curve. The uncalibrated Random Forest curve lies visibly off the diagonal; Platt scaling pulls it back.

The cause is exact and worth stating precisely. On this split the calibration set is *perfectly rank-separated* by the model's scores: every negative scores below every positive. PAVA therefore finds no monotonicity violation to pool, each of the 114 calibration points remains its own block, and each block value equals that point's label — so every fitted value is exactly 0 or 1. Isotonic regression has memorised the calibration labels. This is not a bug: when the monotonicity constraint is never active, the memorising solution *is* the least-squares isotonic solution.

Two clarifications matter. First, the predictions are still piecewise linear, since I interpolate between blocks; what collapses them onto $\{0, 1\}$ is that all block values are themselves 0 or 1, together with flat extrapolation outside the observed range. On this run, 113 of the 114 test predictions come out at exactly 0 or 1. Second, only *two* of those are misclassified, and those two points alone produce the entire jump in log-loss. The magnitude is therefore sensitive to the clipping constant: the same predictions give 0.62 at $\varepsilon = 10^{-15}$ but 0.14 at $\varepsilon = 10^{-3}$. The direction of the effect is robust; the specific number should always be quoted with its ε.

Platt scaling is structurally immune to this failure. Its Bayesian-smoothed targets make it mathematically impossible for the fitted sigmoid to output exactly 0 or 1, no matter how



(a) Diabetes, LR **(b)** Diabetes, RF

Figure 2: Reliability diagrams on the noisier dataset. Predictions spread across the whole $[0, 1]$ range rather than bunching at the extremes, and all three curves stay closer together.

separable the calibration data is. This is a concrete instance of the trade-off in [1]: isotonic regression's flexibility costs it data efficiency, and at these calibration-set sizes the cost dominates.

5 Multi-seed robustness

Every number in Table 2 comes from one random split with 114–154 calibration and test points. That is a single sample from a noisy distribution, not an estimate. I therefore reran the entire pipeline — fresh split, fresh model fit, fresh calibration fit, fresh evaluation — across ten seeds.

Dataset	Model	Uncalibrated	Platt	Isotonic
Breast Cancer	LR	0.085 ± 0.029	0.086 ± 0.027	0.350 ± 0.241
Breast Cancer	RF	0.156 ± 0.111	$\mathbf{0.124 \pm 0.045}$	0.410 ± 0.365
Diabetes	LR	0.487 ± 0.038	0.493 ± 0.035	0.803 ± 0.344
Diabetes	RF	0.481 ± 0.034	0.490 ± 0.034	0.800 ± 0.238

Table 3: Mean test log-loss $\pm$ standard deviation across ten random splits.

Two conclusions follow, and one earlier conclusion is withdrawn.

1. **The isotonic penalty is real and dataset-independent.** Isotonic regression's mean log-loss exceeds Platt's in all four dataset–model combinations, not only in the degenerate Breast Cancer case. The gap is relatively smaller on Diabetes ($\approx 1.6\times$ Platt's, against 3–4$\times$ on Breast Cancer) but it does not vanish.
2. **Its variance is large enough that single-split numbers must not be over-read.** For Random Forest on Breast Cancer the standard deviation (0.365) is nearly as large as the mean (0.410). Notably, Brier score and ECE show no such instability (Figure 3): the effect is specifically an interaction between isotonic's exact-$\{0, 1\}$ outputs and log-loss's unboundedness, not a general claim that isotonic regression is unreliable.
3. **Withdrawn:** on the single reference split it appeared that isotonic regression "has more room to pay off" on the noisier dataset. Ten splits do not support this, and the claim was removed from the analysis notebooks rather than left standing.

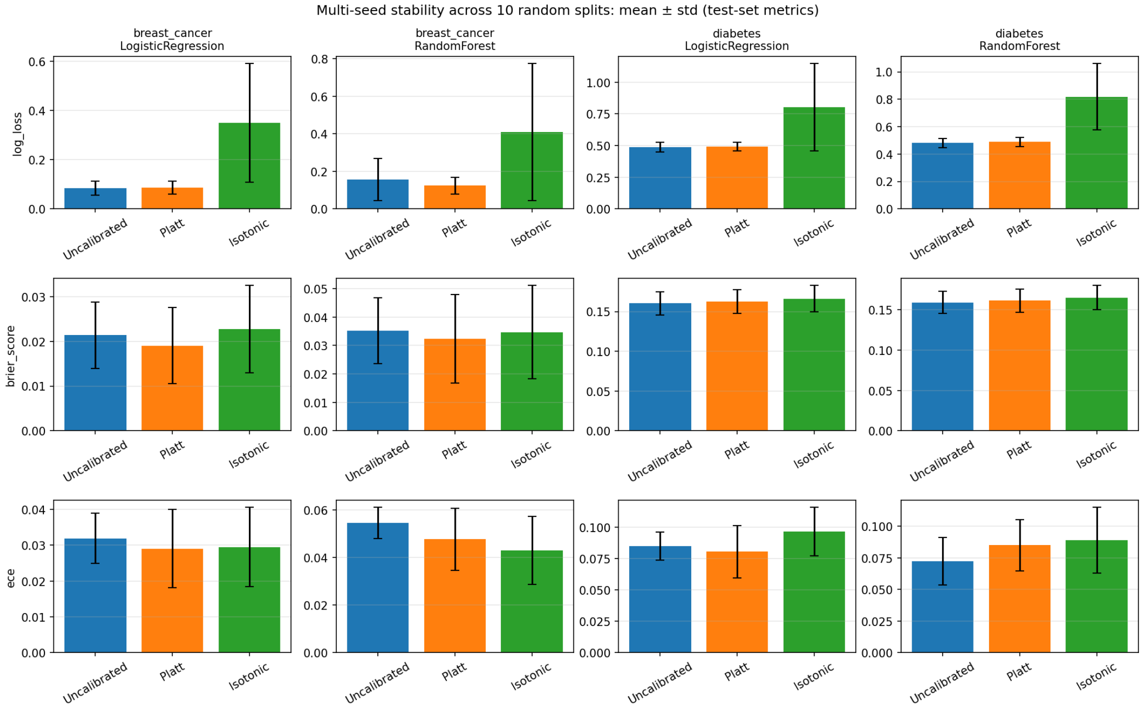


Figure 3: Mean $\pm$ one standard deviation across ten splits. Log-loss (top row) shows isotonic regression's instability clearly; Brier score and ECE, being bounded, do not.

6 Discussion of miscalibration: which model, and in which direction?

Magnitude alone does not answer the question the assignment poses. To identify *direction* I use a signed statistic, the *tilt*: the sample-weighted mean gap (predicted − observed) over the lower half of the probability range, minus the same quantity over the upper half. Positive tilt means predictions are squeezed toward 0.5 (under-confidence, curve flatter than the diagonal); negative tilt means they are pushed away from it (over-confidence).

A plain average of (predicted − observed) across all bins cannot answer this, because the positive gaps at low probabilities and the negative gaps at high probabilities cancel. Verified on synthetic data with a known direction, that naive statistic reads ≈ 0.001 for a strongly under-confident model and ≈ 0.001 for a strongly over-confident one; the tilt separates them cleanly. Direction of miscalibration is a property of the reliability curve's *slope* relative to the diagonal, not of its average offset.

Dataset	**Model**	**Tilt (mean $\pm$ std)**	**Reading**
Breast Cancer	LR	$+0.032 \pm 0.019$	under-confident
Breast Cancer	RF	$\mathbf{+0.059 \pm 0.050}$	under-confident, $\approx 2\times$ more than LR
Diabetes	LR	-0.060 ± 0.115	no stable direction
Diabetes	RF	$+0.031 \pm 0.079$	no stable direction

Table 4: Direction of miscalibration for the uncalibrated models, across ten splits.

On Breast Cancer the prediction of [1] holds: both models are under-confident, and the Random Forest is roughly twice as under-confident as Logistic Regression. This is consistent with the mechanism they describe — averaging many de-correlated trees washes out extreme votes and pulls the ensemble's probabilities toward the centre — a mechanism that applies to the forest and not to the linear model. It is also, tellingly, the one case where Platt scaling clearly helps (log-loss 0.156 → 0.124), which is what one expects when there is a genuine systematic sigmoid-shaped distortion available to undo.

On Diabetes the sign flips from split to split for both models, so no honest claim about direction can be made. Reporting the direction from the single reference split would have produced a confident-sounding statement that ten splits do not support — precisely the error the robustness check exists to catch.

It is also worth noting *where* the miscalibration sits. Per bin, the largest gaps on Breast Cancer occur in the middle of the probability range (roughly 0.2–0.7, gaps up to 0.2–0.3), while the two outermost bins are close to calibrated (gaps of 0.003–0.016). Those outermost bins hold 77–89% of the test mass, which explains why overall ECE stays small (0.03–0.05) despite visible curvature: the models are miscalibrated mostly in the region where they rarely predict.

7 Limitations and negative results

The most important negative result is that, across ten splits, Platt scaling improves ECE in three of four dataset–model combinations, Brier score in two of four, and log-loss in only *one* of four (Random Forest on Breast Cancer). Isotonic regression improves none of them on log-loss. Calibration, in this study, mostly did not help.

This is the expected outcome for these two model families, not a defect in the implementation:

- Logistic Regression is fitted by maximising log-likelihood — it directly optimises a regularised version of the very loss used to evaluate it. On a well-specified problem it is already close to calibrated, so post-hoc correction has almost nothing to correct and mostly adds estimation noise from a calibration set of only 110–150 points.
- Random Forest's vote-averaging *is* a real, correctable distortion, and Platt scaling does correct it — visibly so on Breast Cancer, which is the single combination showing the largest initial miscalibration (ECE 0.0545, the highest of the four).

The deeper limitation is the choice of model families. The dramatic gains in [1] come from boosted trees, SVMs and Naive Bayes — families whose scores are systematically distorted (boosting pushes scores away from 0 and 1 in a characteristic sigmoidal way; SVM margins are not probabilities at all). The assignment specifies Logistic Regression and Random Forest, which sit at the well-calibrated end of that paper's own comparison. This study is therefore structurally unlikely to reproduce the paper's largest effects; its results are consistent with the paper rather than contradicting it. Extending the study with a deliberately miscalibrated base model would be the natural next step.

Two further limitations: ten seeds is enough to detect large effects and expose instability, but not to resolve the small differences between Platt scaling and the uncalibrated baseline; and Random Forest results shift slightly across scikit-learn versions even with a fixed `random_state`, so exact reproduction requires the pinned versions in `requirements.txt`.

8 Conclusion

Both calibration methods were implemented from scratch and behave correctly: Platt scaling agrees with an unregularised logistic regression fitted on the same scores to within 0.005, and the PAVA output is monotonic by construction. The clearest experimental finding is the trade-off between the two methods. Platt scaling's two parameters and smoothed targets make it robust at small calibration-set sizes; isotonic regression's flexibility, valuable in principle, degenerates into label memorisation when the calibration set is small and well separated, producing predictions of exactly 0 or 1 that log-loss punishes severely while accuracy and Brier score barely register. At the sample sizes available here, Platt scaling is the safer default on both datasets.

Equally important is what the multi-seed check revealed: a confident-sounding claim drawn from one split did not survive repetition, and the direction of miscalibration on the noisier dataset turned out not to be identifiable at all. For a study whose entire subject is the trustworthiness of reported probabilities, reporting uncertainty about its own numbers seemed the minimum standard to hold.

References

[1] A. Niculescu-Mizil and R. Caruana. Predicting Good Probabilities with Supervised Learning. *Proceedings of the 22nd International Conference on Machine Learning (ICML)*, 2005.

[2] J. Platt. Probabilistic Outputs for Support Vector Machines and Comparisons to Regularized Likelihood Methods. *Advances in Large Margin Classifiers*, 1999.

[3] H.-T. Lin, C.-J. Lin and R. C. Weng. A Note on Platt's Probabilistic Outputs for Support Vector Machines. *Machine Learning*, 68(3):267–276, 2007.